Where Have All the "Creative Talents" Gone? Employment Dynamics of US Inventors

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Abstract

How are inventors allocated in the US economy and does that allocation affect innovative capacity? To answer these questions, we first build a model where an inventor with a new idea has the possibility to work for an entrant or incumbent firm. Strategic considerations encourage the incumbent to hire the inventor, offering higher wages, and then not implement her idea. We then combine data on 760 thousand U.S. inventors with the LEHD data. We find that when an inventor is hired by an incumbent, their earnings increases by 12.6 percent and their innovative output declines by 6 to 11 percent.

Keyword*:* Inventors, innovation, R&D, firms, dynamism, reallocation

JEL Classification: O3, O4

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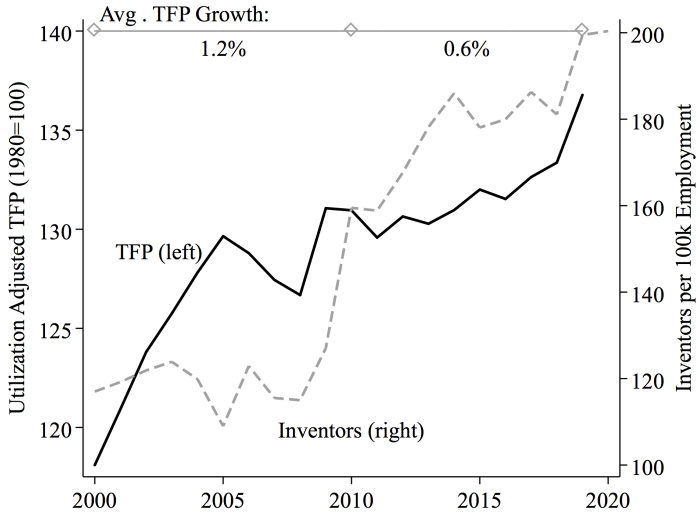
1 Introduction

Societies allocate a significant fraction of their resources into R&D to generate economic growth through technological progress. Many analyses of economic growth focus on innovation inputs, either aggregate R&D spending as a share of GDP or the share of inventors in its workforce (e.g., Jones, 1995). In this paper we argue that it is not only the level of innovation inputs that matters for growth, but also the allocation of those investments. As Arrow (1962) has pointed out, incumbent monopolists suffer from the replacement effect, and may shift their efforts away from producing radical breakthroughs to defensive activities (Akcigit, Baslandze, and Lotti, 2023). As such, the social impact of an inventor may depend on the type of firm they work for.

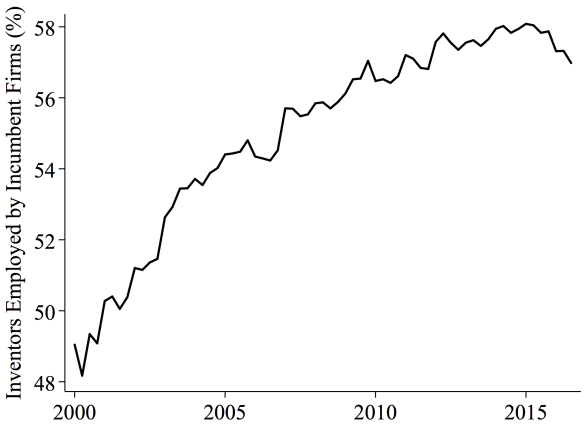
Standard models of endogenous growth (Romer, 1990; Aghion and Howitt, 1992; Grossman and Helpman, 1991; Klette and Kortum, 2004; Akcigit, 2017) predict that economic growth should be proportional to the share of the workforce employed in the research sector. Figure 1 presents seemingly contradictory evidence on this point. Panel A shows utilization-adjusted total factor productivity (TFP) in the U.S. since 2000 (left axis) and a per capita measure of inventor labor (right axis). While there was an acceleration between 2000-2005, after 2005 TFP growth slowed even as the share of inventors grew by over 70%. Innovation inputs are rising as technical progress slows. Equally striking is the shifting allocation of inventors across different-sized firms. Panel B shows that the share of inventors employed by large, incumbent firms rose from 48 percent 2000 to about 57 percent in 2016. Akcigit and Goldschlag (2022) show a complementary fall in the share of inventors employed by young firms. This shifting composition is concerning since inventors at incumbents produce lower quality innovations, shown in Panel C. Panel C shows that inventors at incumbents produce patents with fewer citations, fewer citations per application, fewer independent claims, and more self citations (e.g. more incremental). These striking figures give rise to an important question: How are inventors allocated in the U.S. economy and does that allocation affect innovative capacity?

To answer these questions, we build a tractable creative destruction model that develops intuitions about the strategic incentives incumbent firms face and how they might use innovation input markets to limit competition. In particular, we allow the incumbent to hire an inventor who otherwise would create an innovation inside an entrant firm and displace the incumbent. Since the incumbent monopolist already has a product that it sells to the market, without strategic considerations, it no incentive to innovate and replace its already successful product. However, strategic considerations lead the

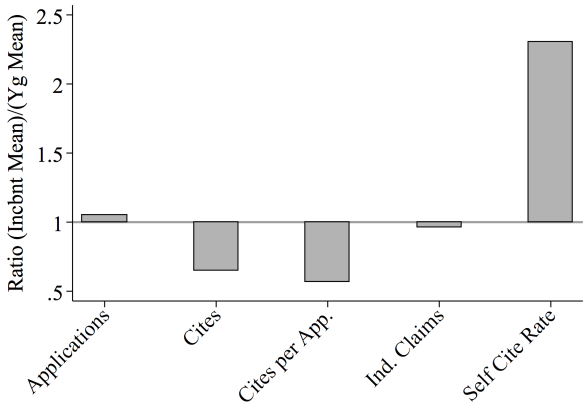
Figure 1: Inventor Employment Dynamics



(a) TFP & Inventors



(b) Inventors at Incumbent Firms



(c) Inventor Productivity, Young & Incumbent Firms

Source: Fernald and Jones (2014), Inventor Employment History, BDS, author's calculations
Notes: Panel A shows utilization adjusted TFP (left axis) indexted to 1980 and the count of unique US-based inventors per 100 thousand employment (right axis). Panel B shows the percent of inventors employed at incumbent firms, which are defined as those $>$ 20 years old and $>$ 1000 employees. Panel C shows the ratio of inventive productivity measures for inventors at incumbent firms relative to young firms, which are defined as firms aged $\leq$ 5 years old. Applications is the count of applications that are eventually granted, cites is the count of citations received, cites per app. is the count of citations divided by the count of applications, ind. claims is the count of independent claims per granted application (Marco, Sarnoff, and deGrazia, 2019), and the self cite rate is the share of citations made to patents with the same assignee.

incumbent to hire the inventor, offering higher wages, and under certain circumstances, choosing not to implement the inventor's idea. The model implies that inventors hired by incumbent firms will earn more but produce fewer innovations compared to inventors hired by entrants.

Our empirical analysis tests the model's implications using data on the employment history of over 760 thousand U.S. inventors. We combine data on individual inventors of patents granted by the U.S. Patent and Trademark Office (USPTO) with information on jobs from the Longitudinal Employer-Household Dynamics (LEHD) Program at the U.S. Census Bureau. Our results show that (i) inventors are increasingly concentrated in large incumbents, less likely to work for young firms, and less likely to become entrepreneurs, (ii) in the cross section, inventors working for incumbent firms earn more and produce less impactful innovations than inventors at young firms, and (iii) when an inventor is hired by an incumbent, compared to a young firm, their earnings increases by 12.6 percent and their innovative output declines by 6 to 11 percent. We also show that these patterns are robust and not driven by promotion to managerial positions in large incumbents, for instance.

Related Literature

Our paper is related to a number of literatures on firms, inventors, and innovation. First, a growing literature has documented that factor reallocation among competing firms is an important source of productivity growth (e.g., Foster, Haltiwanger, and Krizan, 2001, 2006; Acemoglu, Akcigit, Alp, Bloom, and Kerr, 2018). Firms that experience a positive shock, such as the arrival of a new innovation, expand, increasing aggregate productivity. Responsiveness to shocks, however, may depend on market conditions. In recent decades the dispersion of firm-level shocks has risen as firm responsiveness to those shocks as weakened Decker, Haltiwanger, Jarmin, and Miranda (2020). Moreover, models of reallocation typically assume that reallocation, if it occurs, is productivity enhancing. Our work shows that the reallocation of inventor labor may reduce productivity growth when it is the result of strategic behavior by incumbents. We also show how those strategic channels dampen reallocation and limit the replacement of incumbent firms.

Second, a large literature has debated the relative contribution of small, young, and large firms to innovation.[1] Akcigit and Kerr (2018), for example, show that lower quality innovations scale more quickly with firm size. Complementary evidence stresses the

[1]See Akcigit and Kerr (2018) for an overview of this literature.

importance of the strategic behavior of incumbents. Argente, Baslandze, Hanley, and Moreira (2020) show that not only are the patents of large firms less innovative, but they are often used to deter competition. In the extreme, incumbents may acquire entrants to avoid competition and stifle innovation (Cunningham, Ederer, and Ma, 2021). We contribute to this literature by demonstrating that the strategic behavior of incumbents affects how innovation inputs are utilized. Our theoretical framework and empirical evidence is highly complimentary to the work of Cunningham, Ederer, and Ma (2021), who demonstrate the "killer acquisiton" phenomenon both theoretically and empirically. Where they focus on firm-level poaching, our focus is on inventors, employment dynamics, and earnings.

Third, a large literature has documented that capital or labor misallocation can lead to sizable productivity losses (e.g., Hopenhayn and Rogerson, 1993; Hsieh and Klenow, 2009; Hsieh, Hurst, Jones, and Klenow, 2019). In these papers, misallocation is discussed in terms of static, cross sectional misallocation of factor inputs. From a dynamic perspective focused on the supply of inventors, a related literature finds misallocation of talent due to financial frictions, socioeconomic background, and R&D and education policies.[2] Our findings speak to both of these literatures by showing how the effects of misallocation in who becomes an inventor are mediated by how inventors are utilized by firms.

Finally, various recent demonstrated a secular decline in business dynamism in the U.S.. Decker, Haltiwanger, Jarmin, and Miranda (2014) show that the entry rate of firms and establishments has declined over the past several decades. Karabarbounis and Neiman (2014) document the decline in labor share during the same period, while Autor, Dorn, Katz, Patterson, and Van Reenen (2020) argue that market concentration has increased and this increase has been strongly correlated with the decline in labor share at the sectoral level. Akcigit and Ates (2022) provide a quantitative investigation and argue that the decline in knowledge diffusion from market leaders to followers could be an important driver of declining business dynamism. In this paper, we show the rising in concentration of inventors within large incumbents, combined with the strategic hiring of inventors, could lead to a decline in overall dynamism in the U.S. economy.

[2]See Celik (2022), Aghion, Akcigit, Hyytinen, and Toivanen (2017), Bell, Chetty, Jaravel, Petkova, and Van Reenen (2019), Akcigit, Grigsby, and Nicholas (2017), Akcigit, Pearce, and Prato (2020).

2 Model

To motivate and provide a framework for interpreting our empirical analyses, we present a simple model in which incumbents face displacement by innovating entrants. Entrants employ an inventor whose idea can be implemented, at a cost, increasing product quality and capturing the market. Under certain assumptions, incumbents will have an incentive to strategically hire, or poach, the entrant's inventor and not implement the quality-improving innovation. This occurs if implementing the innovation is sufficiently costly relative to the size of the innovative step, making it more profitable for the incumbent to share monopoly rents with the inventor, via higher wages, and avoid displacement by the entrant. In this environment, we expect to see higher wages and less innovative output associated with inventors hired by incumbents relative to inventors hired by entrants.

Production Environment

Individuals consume a unique final good Y_t, which is also used for R&D as discussed below. A measure L units of labor is supplied inelastically by the household. The final good at time t is produced by labor and a continuum of intermediate goods $j \in [0,1]$ with the production technology as follows,

$$Y_t = \frac{L_t^{\beta}}{1-\beta} \int_0^1 q_{jt}^{\beta} k_{jt}^{1-\beta} djt. \tag{1}$$

Henceforth, we will omit the time subscript where doing so does not cause confusion. Producing each unit of k_j has a marginal cost of $\eta > 0$ in terms of the final good. Final good is produced in a perfectly competitive market whereas intermediate goods will be produced by monopolists. Maximization behavior of the final goods producer yields demand for each intermediate good as follows,

$$p_j = L^{\beta} q_j^{\beta} k_j^{-\beta}. \tag{2}$$

This implies that the monopoly profit maximization in sector j can be expressed as follows,

$$\pi_j = \max_{k_j, p_j} \{p_j k_j - \eta k_j\} \quad \textit{Subject to Equation } (2)$$

Hence the equilibrium would be,

$$k_j^* = \left[\frac{1-\beta}{\eta}\right]^{\frac{1}{\beta}} Lq_j \text{ and } p_j^* = \frac{\eta}{1-\beta}.$$

We normalize $\eta = 1 - \beta$ and $L = 1$. Then the equilibrium profit becomes,

$$\pi_j^* = \beta q_j.$$

Innovation Environment

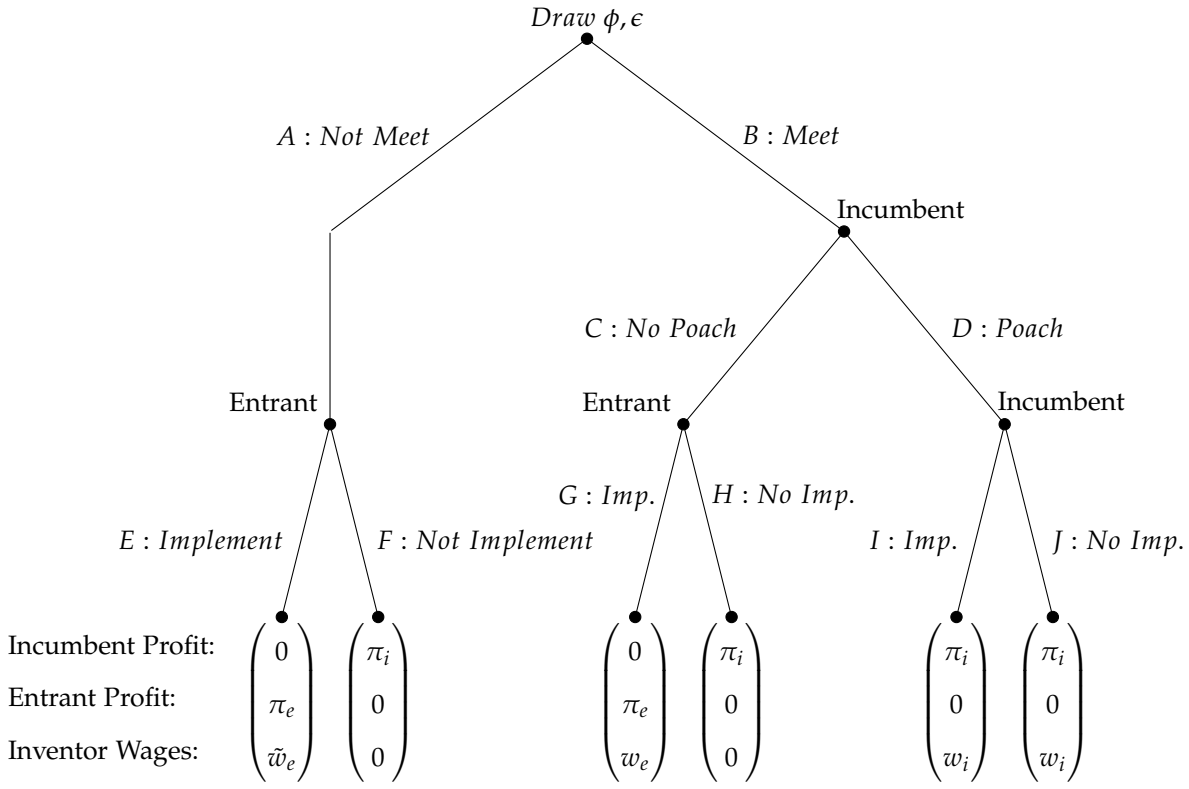
Each monopolist, operating in a given sector, lives for only one period and passes the firm to their offspring. We assume incumbent monopolists maximize their one-period return.[3] Each sector has one potential entrant that employs an inventor. Each inventor has an idea that can be implemented at some cost. In some cases, the incumbent is presented with the opportunity to poach the inventor. Search and matching frictions limit how often the incumbent is able to hire the entrant's inventor. Tightness of the inventor labor market for each sector is fixed, as there is only one inventor, so the incumbent's ability to poach resolves to a simple probability of meeting the inventor. If the entrant implements the inventor's idea, product quality increases and the entrant displaces the incumbent (i.e. creative destruction occurs). When displacing the incumbent, the entrant receives post-innovation profits with upgraded quality net of implementation costs and the incumbent exits. Alternatively, if the incumbent poaches the inventor, the incumbent must choose whether to implement the idea. If the monopoly rents associated with the previous technology dominates the value of implementing the new idea, net of implementation costs, the incumbent will act strategically, poaching the inventor, and not implement the innovation.

More formally, the sequence of events can be described by the game-tree presented in Figure 2, which we describe in greater detail below.

Entrant Innovation. The cost of implementing the inventor's idea is $\phi > 0$. When the inventor's idea is implemented, quality increases from q_j to $(1 + \lambda)q_j$ and the entrant displaces the incumbent. If this occurs, the entrant gets post-innovation profits, with upgraded quality, net of implementation costs, $\beta q_j(1 + \lambda) - \phi$ and the incumbent exits

[3]This relatively simple environment allows us to capture the salient strategic interactions without significant loss of generality.

Figure 2: Innovation Environment Game Tree



Notes: ϕ is the cost of implementing the inventor's idea. ϵ is the probability the incumbent meets the entrant's inventor in the labor market. π_i and π_e are the incumbent and entrant profits, respectively. $\tilde{w}_e$ is inventor wages in the case of no meeting with the incumbent. w_e and w_i are inventor wages at the entrant and incumbent respectively.

and receives zero. We assume that the implementation cost lies between the incremental value of the innovation ($\beta q_j \lambda$) and the post innovation revenues $\beta q_j(1+\lambda)$, as in Inequality 3.

$$\beta q_j(1+\lambda) > \phi > \beta q_j \lambda \tag{3}$$

The remaining possibilities are that $\phi > \beta q_j(1+\lambda) > \beta q_j \lambda$ or $\beta q_j(1+\lambda) > \beta q_j \lambda > \phi$. The former corresponds to no ideas being implemented because they are too costly for both the entrant and incumbent. In the latter case, all ideas will be implemented without strategic considerations because it is always profitable to do so. With Inequality 3, we focus on the most interesting and empirically relevant case.

With probability $\epsilon \in (0,1)$ the incumbent meets the entrant's inventor and has the opportunity to poach. If no meeting occurs, the inventor remains at the entrant and

bargains over wages with no outside options. We will solve the no meeting and meeting cases in turn.

Incumbent does not meet inventor (branch A). With probability $1-\epsilon$ there is no meeting between the incumbent and inventor. In this case the inventor remains at the entrant. Since the cost of implementation is less than the entrant's post-entry return ($\beta q_j(1+\lambda) > \phi$), the entrant will implement the idea and displace the incumbent, following branch AE of Figure 2. We assume Nash bargaining between the entrant and inventor, with the entrant's bargaining power represented by $\gamma \in (0,1)$. Since the entrant and inventor's outside options are zero (branch AE vs AF), the Nash bargaining can be expressed as the following maximization problem.

$$\max_{\tilde{w}_e}(\pi_e - \tilde{w}_e)^{\gamma}\tilde{w}_e^{1-\gamma}$$

The inventor's wages in the case of no meeting with the incumbent is $\tilde{w}_e$ and the entrant's profit is π_e. Through first-order conditions, we find that the inventor's equilibrium wages are a fraction of the entrant's profits equal to their bargaining power $(1-\gamma)$,

$$\tilde{w}_e^* = (1-\gamma)\pi_e$$

Incumbent meets inventor (branch B). If the incumbent meets the inventor it must choose whether or not to poach the inventor. Not poaching the inventor resolves equivalently to not meeting the inventor, described above (A and BC of Figure 2).

The threat of entry creates interesting strategic considerations. Conditional on having poached the inventor, the incumbent's innovation decision (branch BD) takes the following form:

$$\max\left\{\underbrace{0}_{\textit{Value of not implementing}}, \underbrace{\beta q_j \lambda - \phi}_{\textit{Value of implementing}}\right\}$$

The value of not implementing the idea is profits from the existing technology ($\beta q_j > 0$). The value of implementing the idea is post innovation profits net of implementation costs and net of profits associated with the existing technology. Given the assumptions on ϕ in Inequality 3, the value of innovation will be negative, and thus the incumbent chooses not to implement the inventor's idea, following BDJ of Figure 2.

In order to poach the inventor the incumbent must offer wages sufficient to induce the inventor to switch. As with in the no meeting case, we assume that wages are

determined through Nash bargaining with the firm's bargaining power being γ. When bargaining, the inventor's outside option is to work for the other firm in the sector. Conditional on meeting the inventor, the bargaining between the incumbent and the inventor takes the following form:

$$\max_{w_i}(\pi_i - w_i)^{\gamma}(w_i - w_e)^{1-\gamma}$$

Subscripts i and e designate incumbent and entrant values respectively. The incumbent's surplus is profit from the existing technology ($\pi_i = \beta q_j$) net of inventor wages paid by the incumbent (w_i). The inventor's bargaining surplus is the wage at the incumbent (w_i) net of the outside option of wages at the entrant (w_e). A similar, symmetric bargaining problem occurs between the entrant and inventor:

$$\max_{w_e}(\pi_e - w_e)^{\gamma}(w_e - w_i)^{1-\gamma}$$

Now we conjecture, and verify, that $w_i^* > w_e^*$. If incumbent wages are greater than entrant wages, the entrant bargaining problem is maximized when $w_e^* = \pi_e$. Intuitively, if the incumbent offers higher wages, the inventor can always do better by working for the incumbent. Since the entrant's outside option is not entering and receiving zero, the entrant is willing to set the inventor's wages equal to its profit. Solving the incumbent bargaining problem, given entrant wages are equal to entrant profit, the inventor's equilibrium wages at the incumbent can be written as follows:

$$w_i^* = (1-\gamma)\pi_i + \gamma\pi_e$$

Our assumption in Inequality 3 implies that $\pi_i > \pi_e$ since $\pi_i = \beta q_j$ and $\pi_e = \beta q_j + (\beta q_j \lambda - \phi)$ and $\phi > \beta q_j \lambda$. As $\pi_i > \pi_e$ and $w_i^* > w_e^*$, our conjecture is verified. Note that since entrant wages are equal to entrant profits the model also captures the dynamics of inventor entrepreneurship. If the inventor owns the entrant, rather than being employed by it, the inventor's return remains equal to post-entry profits and strategic interaction with the incumbent remains unchanged.

The model yields the following predictions. When an incumbent producing using an older technology has a cost advantage over an innovative entrant, the model predicts:

Prediction 1: The incumbent will poach the inventor and not implement the inventor's idea, reducing aggregate innovative activity.

Prediction 2: The incumbent will poach inventor entrepreneurs and not implement the inventor's idea, reducing aggregate innovative activity.

Prediction 3: The incumbent will offer higher wages than the entrant in order to poach the inventor from the entrant.

3 Empirical Evidence

This section describes the empirical results on inventor productivity and firm characteristics.

Data

Patent Data. We use data from the USPTO on patents granted between 2000 to 2019 to identify inventors of granted patents and characterize their inventive output each period. Information on patent grant and application dates are used to create counts of patents associated with each inventor each period. We use data on citations received to citation weight patent counts and count self citations. We also utilize claim counts, as well as the categorization of claims as either independent or dependent, developed by Marco, Sarnoff, and deGrazia (2019).

Data Linking. We use linkages between inventor records (patent, inventor sequence combinations) and the Census Bureau's disambiguated, anonymized, person-level identifiers, known as Protected Identification Keys (PIKs), developed and described by Akcigit and Goldschlag (2022). With PIKs are associated with patent inventor records, we are able to match inventors to survey, census, and administrative Census Bureau micro data on individuals. Ultimately, we observe the employment histories of approximately 760 thousand inventors associated with 3.6 million patents granted.

Administrative and Survey Data. We develop our Inventor Employment History database using data from the LEHD Program at the U.S. Census Bureau. LEHD contains quarterly job-level observations drawn from administrative state-level unemployment insurance records. We restrict to 45 states with employment data starting in 2004 or earlier. Our job-level panel includes only dominant (highest earnings), beginning-of-quarter jobs from 2000Q1 to 2016Q3.[4] Our final sample contains nearly 35 million quarterly job observations.

We augment LEHD employer information with firm characteristics from the Longitudinal Business Database (LBD) (Haltiwanger, Hyatt, McEntarfer, Sousa, and Tibbets,

[4] A beginning-of-quarter job is defined as a job for which the individual has positive earnings in both t and $t-1$. We select the dominant, or highest earning beginning-of-quarter job. Some states enter the data after 2000Q1.

2014). Finally, we collect information on occupation from the American Community Survey (ACS). Despite being relatively rare—the ACS sample ranges from 800 thousand to 3.5 million housing units during our sample period—we are able to attach occupation information to over 100 thousand of our inventor-job observations.

Measures

Inventive Productivity. We count the number of granted patent applications ($Apps$) and applications weighted by the number of citations received in the first five years after the patent was granted ($Cites$). Windowing citation counts limits truncation effects due to older patents having more time to accrue citations. We also compute the number of citations received per application in the period ($Cites\ Per\ App$). We use the count of citations made and self citations, those where the citing and cited patent have the same assignee, to compute the share of self citations ($Self\ Cite\ Rate$). Patents that primarily cite other patents owned by the same firm are more incremental in nature (Akcigit and Kerr, 2018). The scope of patents is measured using the share of claims that are classified as independent ($Independent\ Claims$) using data developed by Marco, Sarnoff, and deGrazia (2019).

Employment Outcomes. Our measure of quarterly earnings is the natural log of real full-quarter earnings ($ln(Earnings)$).[5] Hire and separation events are identified using the sequence of employers observed in an individual's employment history.[6] Following Choi, Goldschlag, Haltiwanger, and Kim (2021), we identify entrepreneurs as individuals that receives earnings from a startup "on day one" and are among the top earning workers at the firm in the first year.

To better approximate the intuitions of the model, we classify employers based on size and age as either a young or incumbent firm. Young firms are those less than age six. Incumbents are older larger firms that are at least 21 years old and have at least 1,000 employees. This dichotomy sharpens the contrast in the types of firms that employ inventors.

[5]Full quarter earnings is only observed when a job has positive earnings in both the previous and subsequent quarters. This reduces the likelihood that an earnings measure represents only partial quarter employment. Earnings is normalized to 2012q1 dollars and demeaned within 6-digit NAICS and quarter.

[6]Stable hire and separations are defined analogously to those measured in the Quarterly Workforce Indicators (QWI).

Results

In our Inventor Employment History database roughly 12 percent of inventors are female, 27 percent were born outside of the US, and approximately 9 percent are observed starting at least one business during our sample period.[7] Over 60 percent of the inventor-quarter observations are of inventors aged 36 to 55, making our inventor sample older, on average, than the US workforce.[8] Approximately 55 percent of inventor jobs are at incumbent firms while about 10 percent are at young firms. Comparing this to data from the Census Bureau's Business Dynamics Statistics, approximately 38 percent of all employment during this period was associated with incumbent firms and 12 percent in young firms.[9] The hire and separation rates we observe for inventors are about 4.6 percent and 4.4 percent, respectively.[10]

We find interesting time series patterns for several of our employment dynamics measures. As shown in Figure 1, the share of inventors employed by incumbents rose by 17%, or 8 percentage points, from 48.9 percent in 2000 to 57.3 percent in 2016. Also, as shown by Akcigit and Goldschlag (2022), hire and separation rates for inventors are declining over this period. The hire and separation rates of inventors fell by roughly 40% between 2000 and 2016, from about 7 percent to less than 4 percent. Meanwhile, workers in similar industries saw hire and separation rates that were fairly stable at 6 to 6.5 percent, suggestive of declining employment dynamism among inventors.

The entrepreneurship rate among inventors also fell over this period. Panel A of Figure 3 shows the change in the likelihood that an inventor starts a new business each year, relative to 2000, after controlling for individual fixed effects. Entrepreneurship among inventors is relatively rare. Though about 9 percent of inventors are *ever* observed as entrepreneurs, only 0.639 percent founded firms in 2000. Since 2000, the inventor entrepreneurship rate fell by roughly 44%, or nearly 0.3 percentage points.[11] This is especially concerning since firms founded by inventors have a much steeper life cycle curve. Panel B of Figure 3 shows the change in firm size over the firm's life cycle for inventor founded and non-inventor founded firms, relative to its size at birth, after controlling for

[7] Akcigit and Goldschlag (2022) report the foreign born share of *grant active* inventors to be higher at about 30 percent over the period, rising from 23 percent to 34 percent over this period.

[8] In 2010, approximately 46 percent of the US labor force was between the ages of 35 and 54. See the Bureau of Labor Statistics Current Population Survey Employment status of the civilian noninstitutional population by age, sex, and race tables for details https://www.bls.gov/cps/aa2010/aat3.txt.

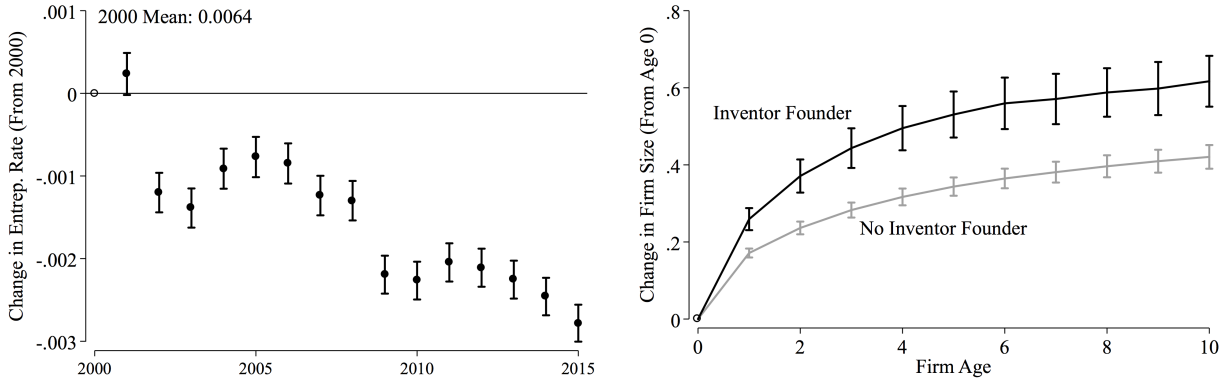
[9] See the Business Dynamics Statistics tables for details https://www.census.gov/data/datasets/time-series/econ/bds/bds-datasets.html.

[10] See Akcigit and Goldschlag (2022) for additional descriptive analyses of the Inventor Employment History database.

[11] Akcigit and Goldschlag (2022) show that the decline in entrepreneurship rates was particularly large for super star inventors, those with the most highly cited patents.

industry-year effects. Conditional on survival, by age five inventor founded firms grow by nearly 70 percent while non-inventor founded firms grow by 41 percent.[12] Not only do inventor founded firms experience higher growth rates, they are also larger at birth. On average, inventor founded firms have 8.1 employees and non-inventor founded firms have 5.7 employees.[13] The results in Figure 3 suggest that the high growth young firms generated by inventors are becoming more rare over time.

Figure 3: Inventor Entrepreneurship



(a) Declining Inventor Entrepreneurship (b) Inventor Firm Performance

Source: Inventor Employment History, author's calculations
Notes: Panel A reports estimates of β_ks from the following equation, $Entrep_{i,t} = \alpha + \sum_{k=2001}^{2015} \beta_k D_t^k + \psi_i + \epsilon_{i,t}$. For inventor i in year t, $Entrep_{i,t}$ equals 1 if the inventor started a business in that year and zero otherwise, D_t are year effects, and ψ_i are person fixed effects. The mean of $Entrep_{i,t}$ in 2000 is 0.0064. Panel B shows estimates of β_as from the following equation, $ln(FirmSize_{f,t}) = \alpha + \sum_{a=1}^{10} \beta_a FirmAge_{f,t} + \delta_{f,t} + \epsilon_{f,t}$. For firm f in year t, $ln(FirmSize_{f,t})$ is the log of firm employment in year t, $FirmAge_{f,t}$ is the firm's age in year t, and $\delta_{f,t}$ are industry-year fixed effects.

Though many measures of dynamism fell over this period, one might have expected those trends to differ for the high human capital inventive workforce. Instead, our data suggests inventors are less likely to change jobs, less likely to start a business, and increasingly concentrated among older, larger incumbent firms. In the cross section, inventors employed by incumbents produce patents that have fewer citations, fewer independent claims, and more self citations (Panel C of Figure 1). Moreover, inventors employed by incumbents have 13% higher earnings than inventors at young firms.[14] A

[12] Elasticities for $ln(FirmSize)$ are interpreted using the $e^{\beta_a} - 1$, where β_a is defined as in the equation in the Figure notes.

[13] Standard errors for these means are (32.5) and (24.1), respectively.

[14] Additional summary statistics on the Inventor Employment History database can be found in the

back-of-the-envelope calculation suggests that the shift of inventors towards incumbents is associated with, ceteris paribus, a 5.3% decrease in citations.[15]

To more precisely estimate differences in inventor output and earnings between inventors at young and incumbent firms, we propose a matching exercise and event study framework. We identify similar inventors hired by incumbents and young firms and measure how their outcomes differ after the hire. We identify young and incumbent firm hire events and create an annual panel of earnings and inventive output measures before and after the hire.[16] To find "pairs" of very similar hire events, we classify inventors into deciles of the distribution of inventive output or earnings in each year prior to the hire. For each outcome measure, we match on these decile bins, sector, and hire quarter. When more than one potential match exists, we select a single match that is closest in age and pre-hire inventive productivity or earnings.[17] Incorporating age in our matching algorithms ensures that differences in earnings and inventive output do not reflect inventor life cycle effects. Our matching strategy generates very similar pre-hire outcomes (see bottom panel of Table 1).

With the sample of matched incumbent and young firm hires, we measure the difference in post-hire outcomes between the two groups by estimating the following equation. For inventor i, in year relative to the hire t, and hire event e we estimate,

$$Y_{ite} = \alpha + \sum_{j=-4}^{4} \lambda_j d[j]_{ite} + \beta_1 Incumbent_{ie} + \sum_{j=-4}^{4} \eta_j d[j]_{ite} \times Incumbent_{ie} + \beta_2 Age_{ite} + \delta_j + \gamma_k + \psi_i + \epsilon_{ite} \tag{4}$$

$Incumbent_{ie}$ is equal to one if the hire event occurred at a incumbent firm, zero if it occurred at a young firm. δ_j, γ_k, and ψ_i are sector, hire year, and person fixed effects respectively. Person-level fixed effects will control for time invariant characteristics of

Online Appendix.

[15]To compute this we use the young and incumbent share in 2000 and 2016, normalized to ignore the excluded group, then compute the change in innovative output, given their cross sectional mean, due to shifting composition towards incumbents.

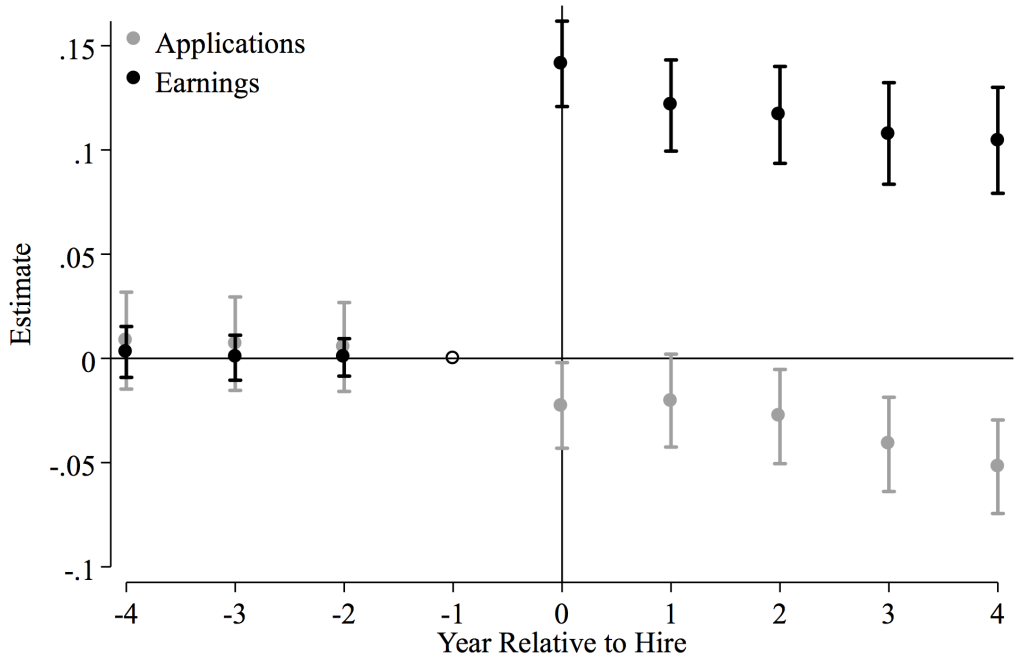
[16]For the event study we aggregate from the quarterly observations in our inventor jobs panel to the annual level, summing applications and citation weighted applications and taking the mean of full quarter earnings, self citation rate, independent claim share, and citations per application. Individuals with no applications in a given year will appear with zero applications while those with no full quarter earnings will be have a missing earnings value for the year (and are therefore dropped).

[17]We minimize the normalized euclidean distance between age and inventor output or earnings in each year $t-1$ to $t-4$. A young firm hire event cannot match to multiple incumbent hire events and vice versa. An inventor may experiences multiple hire events. For additional details and match balance statistics seethe Online Appendix.

individual inventors. Age_{ite} is the inventor age. λ_js capture year relative to hire effects. Our focus will be estimates of the η_js, which capture the difference in outcome Y between inventors hired by incumbents to similar inventors hired by young firms in each period before and after the hire event.

Figure 4 shows estimates from Equation 4 for earnings and applications. Our matching, by construction, eliminated pre-event differences. In the year of the hire event ($t = 0$) incumbent hires have 15 percent higher earnings than young firm hires.[18] This stabilizes to roughly 11 percent four years after the hire event. At the same time, incumbent hires see a 0.023 lower application count which falls to a 0.052 gap four years after the hire event. A decline in applications of 0.052 represents approximately 8 percent of the pre-hire mean application count (see bottom panel of Table 1).

Figure 4: Young and Incumbent Inventor Hires



Source: Inventor Employment History, Founding Team Database
Notes: Shown are event study estimates for years relative to the hire event of the difference in an inventor's outcome, patent applications and log earnings, between inventors hired by an incumbent firm and those hired by a young firm. We match inventor hire events at young firms (those age ≤ 5) and incumbent firms (firms at least 21 years old and with at least 1,000 employees). Matching is performed on hire period and pre-hire period characteristics. Shown are estimates of η_j from Equation 4.

[18] Elasticities for $ln(Earnings)$ are interpreted using the $e^{\eta_j} - 1$.

As noted previously, we consider several different measures of inventive output. To explore these alternative measures, we collapse the time dimension of our event study into a difference-in-differences framework with a $Post_{ite}$ binary that is equal to one if $t \geq 0$ and zero otherwise. The estimation equation becomes the following.

$$\begin{aligned} Y_{ite} = \alpha + \ & + v_1 Post_{ite} + v_2 Incumbent_{ie} + v_3 Post_{ite} \times Incumbent_{ie} \\ & + \beta_4 Age_{ite} + \delta_j + \gamma_k + \psi_i + \epsilon_{ite} \end{aligned} \tag{5}$$

The difference-in-differences estimates for for v_1 and v_3 along with associated pre-hire means are presented in Table 1. In the first column for $Apps$ we find 0.036 lower applications for incumbent hires in the post period, roughly 6 percent of the pre-hire mean for those hired by incumbents. For earnings, we find 12.6 percent higher post-hire earnings for inventors hired by incumbents. Across the other measures, incumbent hires have 11.9 percent lower citations, 11.3 percent lower citations per application, 5.4 percent lower independent claim share, and 37.7 percent higher self citation rates, all relative to pre-hire incumbent means.

Table 1: Young and Incumbent Hires: Alternative Outcomes

	Apps	$ln(Earnings)$	Citations	Citations per App	Independent Claims	Self Cite Rate
Post	-0.1289***	-0.03429***	-0.8955***	-0.2611***	-0.07455***	-0.01234***
	(0.006064)	(0.007612)	(0.2094)	(0.02871)	(0.002297)	(0.001006)
Post x Incumbent	-0.03585***	0.1184***	-0.6861***	-0.1145***	-0.01581***	0.02278***
	(0.007011)	(0.009772)	(0.2069)	(0.02439)	(0.002392)	(0.001358)
R^2	0.311	0.8106	0.2531	0.1845	0.1818	0.4716
N	781,000	143,000	371,000	370,000	660,000	213,000
Pre-Hire Means						
Incumbent Hires	0.6017	10.29	5.758	1.009	0.2925	0.06035
	(1.34)	(1.045)	(33.01)	(4.246)	(0.5645)	(0.132)
Young Hires	0.6068	10.29	5.933	1.039	0.2947	0.0601
	(1.429)	(1.051)	(33.27)	(4.338)	(0.5795)	(0.1309)

Source: Inventor Employment History, author's calculation
Notes: Difference-in-Differences estimates for the hire event for each outcome $Y_{i,t,e}$, between inventors hired by an incumbent firm and those hired by a young firm. Estimates are of v_1 and v_3 from Equation 5. Standard errors are shown in parentheses. Observation counts vary by measure because outcome relies a separate match, minimizing the difference in pre-match values of the given measure. Incumbent binary included but not reported. Observation counts have been rounded to avoid the disclosure of confidential information.

The estimates shown in Table 1, though not casual, are in agreement with the cross sectional differences in earnings and inventive output referenced earlier and are consistent with the predictions of the model. Taken together these estimates suggests inventors employed by or hired by incumbent firms, relative to young firms, have higher earnings and lower inventive output.

Robustness

In this section we demonstrate the robustness of our estimates to several concerns about measurement and mechanisms.

Stayers and Leavers. Our panel does not condition on inventors remaining at the hiring firm. Many inventors in our sample change jobs after the hire event. About 40% of incumbent hires and 33% of young firm hires are still at the firm five years after the hire event ($t+4$). Despite this, time trends in the share of inventors remaining at the hire firm are similar between incumbent and young firms. Moreover, we find qualitatively and quantitatively similar effects when we subset to observations for which the the inventors remain at the hiring firm.[19]

Mechanical Self Citation Rate Effects. One potential concern with the self citation rate estimates is a mechanical relationship driven by incumbents having larger patent portfolios. To determine the extent of this mechanical effect we create a simulated self citation rate by randomly generating self citations in proportion to the assignee's share of the technology groups cited by the patent. Estimating Equation 5 with this simulated self citation rate we find estimates that are an order of magnitude lower. The estimate for $Post \times Incumbent$ is 0.002107 compared to 0.02278 found in Table 1, suggesting that self citation rates are not primarily driven by this mechanical effect.[20]

Occupational Differences. A potential explanation for the event study patterns is that inventors at incumbents are more likely to be managers than engineers. In the ACS-matched sample, 49.8% of inventors are in technical occupations occupations while 26% are in managerial occupations.[21] As for young and incumbent firms, inventors at young

[19] See Online Appendix for estimation output.

[20] The standard error for the simulated self citation rate $Post \times Incumbent$ estimate is (0.0003031). See the Online Appendix for additional estimation details.

[21] Technical occupations include Computer and Mathematical, Architecture and Engineering, or Life, Physical, and Social Science occupations, which correspond to ACS occupation codes 1000 to 1965. Management occupations include Management, Business Operations, or Financial Specialists, which correspond to ACS occupation codes ≤ 950.

firms are more likely to be managers (31% vs. 23%) and less likely to be in technical occupations (44% vs. 57%) than inventors at incumbent firms. These patterns are intuitive if we think inventors at young firms tend to be key employees or founders that bring both innovative and managerial skills to the firm. These shares suggest that the diverging earnings and output measures we find are not driven by differences in occupation.

Adjustment Costs at Incumbents. Another potential explanation consistent with our findings is large post-hire adjustment costs for inventors hired by incumbents, who may be less flexible due to additional layers of bureaucracy (Williamson, 1984) or have established R&D projects that new inventors must integrate with. To test this possibility, we define a new measure that captures whether an inventor has a patent application in a technology class that is new to the inventor in a given year. The measure, $NewCPC_{it}$, is equal to 1 if an inventor has a patent application at time t in a major CPC class that the inventor has no prior applications in and zero otherwise. Estimation results suggest that incumbent hires are actually less likely to submit patent applications in a new-to-the-inventor CPC class than young firm hires.[22]

Hire Events in Great Recession. Our hire event study requires information on earnings and patenting activity before and after the hire event, which limits our hire events to years 2005 to 2012, straddling the great recession. To ensure our results are not driven by the great recession years, we estimate our difference-in-differences specification with hire events occurring in 2005 and 2006, again finding negative effects for citations per application and positive effects for earnings.

4 Conclusion

Talent is the most important ingredient for innovation and the allocation of talent across firms has a first-order impact on the innovative capacity of an economy. In this paper, we develop a theory of creative destruction that focuses on the allocation of inventors across firms. Our model describes the strategic incentives faced by incumbents. To avoid displacement by the entrant, incumbent firms will poach inventors and shelve their innovations.

On the empirical side, our paper uses novel data on the employment history of over 760 thousand inventors to investigate the model's predictions. We show empirically that

[22]See Online Appendix for estimation output.

(i) inventors are increasingly concentrated in large incumbents, less likely to work for young firms, and less likely to become entrepreneurs, and (ii) when an inventor is hired by an incumbent, compared to a young firm, their earnings increases by 12.6 percent and their innovative output declines by 6 to 11 percent.

These findings have very important policy implications. First, country-level aggregates of innovation inputs (e.g. R&D spending or inventors per capita) may be misleading, failing to provide a complete picture of a country's innovative capacity. While a higher share of inventors is desirable, the composition of those inventors across firm types is also important. Second, not all factor reallocation is necessarily good. Our analysis documents that inventor reallocation toward large incumbents, at least the way it happened in recent decades, may be lowering the growth capacity of the country. Finally, policies that encourage more incumbent innovation might occur at the expense of entrant innovations, which are higher quality on average.

These important findings point to several interesting, policy-relevant questions. For instance, what role do non-compete agreements play in explaining when inventors work for incumbents or young firms? Those policies may play an important role in generating spin-offs and inventor entrepreneurship. In addition, what role do financial frictions play in explaining inventor's choice to work for incumbents? If an inventor faces credit constraints, that may weaken incentives to work for an entrant or start a new firm. We hope that the data developed by Akcigit and Goldschlag (2022) will spur further research on these topics and shed light on the "black box" of inventor employment dynamics in the US.

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Online Appendix for

Where Have All the "Creative Talents" Gone? Employment Dynamics of US Inventors

– Not for Publication unless Requested –

Ufuk Akcigit Nathan Goldschlag

A.1 Inventor History Summary Statistics

Table A1 provides summary statistics for our Inventor Employment History database. The Panel A provides characteristics and employment dynamics measures and the bottom two panels (B and C) summarize our earnings and inventive outcome measures for inventor jobs at incumbent and young firms respectively. Observation counts are reported in thousands. The observation counts for the first three rows are the count of unique inventors. In the remaining rows, observation counts represent the number of quarterly job observations. For example, for Female in Table A1, there are 760 thousand inventors, 11.7 percent of which are female. In contrast, for Age $\leq$25, there are nearly 34.7 million job observations, 3.4 percent of which cover inventors inventors age less than or equal to 25 years old.

TABLE A1: SUMMARY STATISTICS

	Mean	Std. Dev	N (000s)
Panel A: Characteristics and Dynamics			
Female	0.1168	0.3212	760
Foreign Born	0.2686	0.4432	760
Ever Entrepreneur	0.0896	0.2856	759
Age ≤ 25	0.03352	0.18	34,690
Age 26-35	0.2051	0.4038	34,690
Age 36-45	0.3359	0.4723	34,690
Age 46-55	0.2994	0.458	34,690
Age ≥ 56	0.126	0.3319	34,690
Incumbent Employer	0.551	0.4974	34,690
Young Employer	0.1027	0.3036	34,690
Hires	0.04558	0.2086	34,570
Separations	0.04442	0.206	34,570
Panel B: Incumbent Employer			
ln(Earnings)	10.33	0.659	18,320
Apps	0.09188	0.5301	19,110
Cites	0.5371	22.29	15,920
Ind Claim per App	0.1575	0.7996	17,110
Cite per App	4.852	15.44	1,103
Self Cite Rate	0.1429	0.193	1,176
Panel C: Young Employer			
ln(Earnings)	10.2	0.8675	3,262
Apps	0.08717	0.4543	3,563
Cites	0.8275	10.09	3,145
Ind Claim per App	0.1634	0.9009	3,300
Cite per App	8.511	19.91	202
Self Cite Rate	0.06212	0.1289	213

Source: Inventor Employment History, Founding Team Database
Notes: Observation counts have been rounded to avoid the disclosure of confidential information.

Hire Event Matching

This analysis measures the impact on an inventor's earnings and inventive output of being hired by an incumbent firm *relative to* being hired by a young firm. To do this, we identify all hire events for inventors in which the hiring firm is either an incumbent or a young firm. With those hire events we construct a balanced panel of annual observations before and after the hire. The panel contains 9 years, 4 years before the event, $t = 0$ is hire event quarter and the following 3 quarters, and then 4 years following the hire year. We track measures of inventive productivity (apps, citations, etc.) and earnings aggregated

to the annual level for each year (lead and lag) relative to the hire event. We aggregate to annual measures of earnings and inventive productivity by summing applications and citation weighted applications and taking the mean of full quarter earnings, self citation rate, independent claim share, and citations per application. Individuals with no applications in a given year will appear with zero applications while those with no full quarter earnings will be have a missing earnings value for the year (and are therefore dropped). Note that if an inventor has multiple back-to-back hire events, the relative year measures will overlap for the different inventor-hire event combinations.

With this inventor-hire event-relative year panel we perform our matching, aiming to find incumbent and young hire event twins—young and incumbent hire events in which the inventors have very similar pre-hire inventive productivity or earnings. We perform a separate matching exercise for each of our outcome variables (applications, earnings, etc.). We create percentile bins (10 for patent measures and 50 for earnings) of the outcome variable by relative year prior to hire (e.g. separate bins for $t-4$, $t-3$, etc.). Call these percentile bins p_{t-4}, p_{t-3}, p_{t-2}, and p_{t-1}. Similarly, call the outcome measure in each relative year y_{t-4}, y_{t-3}, etc.

Records with a zero measure in the outcome variable are grouped together within a relative year, which yields 11 bins for cite measures and 51 for earnings. Inventor-hire event pairs are dropped if (1) all outcome measures are either missing or zero for all pre-event relative years. These "all zero" cases are events we know relatively little about–it is difficult to assess the quality of an inventor or their trajectory if we observe no inventive activity within our window around the hire event.

The matching iterates over each young firm hire event, block matching to all incumbent hire events by [year and quarter of hire, sector,p_{t-4},p_{t-3},p_{t-2},p_{t-1}]. In many cases, this yields more than one potential match for a given young hire event. To resolve this, we calculate the euclidean distance, $dist_{i,j}$, between the outcomes in each prior year and age. For young hire event i and incumbent hire event j, the distance is defined as follows.

$$dist_{i,j} = \sqrt{yDiff^2_{t-4,i,j} + ... + yDiff^2_{t-1,i,j} + ageDiff^2_{i,j}} \tag{6}$$

The difference in each prior year, $yDiff_{t-k,i,j}$, is noramalized to fall between 0 and 1, and is defined as follows.

$$yDiff_{t-k,i,j} = \frac{abs(y_{t-k,i} - y_{t-k,j}) - \min_{\forall j}(abs(y_{t-k,i} - y_{t-k,j}))}{\max_{\forall j}(abs(y_{t-k,i} - y_{t-k,j})) - \min_{\forall j}(abs(y_{t-k,i} - y_{t-k,j}))} \tag{7}$$

The normalized difference in age, $ageDiff_{i,j}$, is defined analogously to $yDiff_{t-k,i,j}$. The matching algorithm minimizes the $dist_{i,j}$, breaking ties randomly. We match without replacement—once an incumbent hire event has been matched to a young hire event it cannot match to any other young hire event.

Table A2 shows the mean and standard deviation of pre-hire outcomes and age for the matched young and incumbent hires along with a ttest for the difference in means. Some differences in age are statistically significant, but all are less than one year. None of the differences in pre-hire outcomes are statistically significantly different.

Table A2: Hire Event Match Balance

Variable	Mean Young	Mean Incumbent	Diff
Applications			
Age	40.3	40.26	-0.04
	(9.613)	(9.602)	(0.03262)
Apps	0.6068	0.6017	-0.0051
	(1.429)	(1.34)	(0.004703)
Citations			
Age	39.93	39.61	-0.32
	(9.543)	(9.309)	(0.04644)
Cites	5.933	5.758	-0.175
	(33.27)	(33.01)	(0.1633)
Independent Claim Share			
Age	40.26	40.22	-0.04
	(9.636)	(9.556)	(0.03499)
Ind Claim Share	0.2947	0.2925	-0.0022
	(.5795)	(.5645)	(0.002086)
Cites per Application			
Age	39.91	39.57	-0.34
	(9.551)	(9.285)	(0.04648)
Cites per App	1.039	1.009	-0.03
	(4.338)	(4.246)	(0.02119)
Self Cite Rate			
Age	38.61	37.98	-0.63
	(10.54)	(10.73)	(0.02219)
Self Cite Rate	0.0601	0.06035	0.00025
	(.1309)	(.132)	(0.0008218)
$ln(Earnings)$			
Age	40.18	40.05	-0.13
	(9.48)	(9.505)	(0.07345)
ln(Earnings)	10.29	10.29	0
	(1.051)	(1.045)	(0.008108)

Source: Inventor Employment History, Founding Team Database
Notes: Match balance statistics for pre-hire characteristics from hire event study. Standard errors are shown in parentheses.

Simulated of Self Citation Rates

Table A3 shows estimation results of Equation 5 where Y_{ite} is a simulated self citation rate in which, given the number of citations a patent makes to each CPC technology

group, a patent cite's the assignee's existing patent corpus in proportion to it's share of all patents in that CPC group.

Table A3: Simulated Self Citation Rates

	Self Cite Rate Simulated
Post	-0.001016***
	(0.0002248)
Post x Incumbent	0.002107***
	(0.0003031)
R^2	0.3374
N	213,000

Source: Inventor Employment History, author's calculation
Notes: Estimates are of v_1 and v_3 from Equation 5. Standard errors are shown in parentheses. Incumbent binary included but not reported. Observation counts have been rounded to avoid the disclosure of confidential information.

New to Inventor Major CPC

Table A4: New Major CPC

	New CPC
Post	-0.2013***
	(0.001729)
Post x Incumbent	-0.002968**
	(0.001037)
R^2	0.1499
N	548,000

Source: Inventor Employment History, author's calculation
Notes: Estimates are of v_1 and v_3 from Equation 5. Standard errors are shown in parentheses. Incumbent binary included but not reported. Observation counts have been rounded to avoid the disclosure of confidential information. New CPC is a binary equal to one if the inventor has an application in a major CPC technology group that the inventor is not previous observed with patent applications in that major CPC. This measure is left censored in 2000, the beginning of our data.

Age and Life Cycle Effects

Table A5: Age of Inventor Heterogeneous Effects

	Apps	$ln(Earnings)$	Citations	Citations	Independent	Self Cite Rate
Post	-0.0875***	0.1425***	-0.7108**	-0.285***	-0.06288***	-0.01155***
	(0.006729)	(0.005931)	(0.2374)	(0.03404)	(0.002914)	(0.001388)
Post × Incmbnt	-0.03515***	0.1311***	-0.7268**	-0.112**	-0.01578***	0.02411***
	(0.007388)	(0.006376)	(0.2307)	(0.0343)	(0.003255)	(0.001869)
Post × Incmbnt × GT Median Age	-0.002157	-0.02288**	0.07252	-0.003854	-0.0001692	-0.002262
	(0.01031)	(0.008913)	(0.3262)	(0.04415)	(0.004564)	(0.00257)
R^2	0.3794	0.8408	0.3308	0.2694	0.2657	0.6357
N	781,000	143,000	371,000	370,000	660,000	213,000

Source: Inventor Employment History, author's calculation
Notes: Estimates for ν_1, ν_4, and ν_6 from the following equation.

$$
\begin{aligned}
Y_{ite} = \alpha + \ & + \nu_1 Post_{ite} + \nu_2 Incumbent_{ie} + \nu_3 GTMed_{it} \\
& + \nu_4 Post_{ite} \times Incumbent_{ie} + \nu_5 GTMed_{it} \times Post_{ite} \\
& + \nu_6 GTMed_{it} \times Post_{ite} \times Incumbent_{ie} + \\
& + \beta_4 Age_{ite} + \delta_j + \gamma_k + \psi_i + \epsilon_{ite}
\end{aligned}
$$

Table A6: Hire Effects for Stayers

	Apps	$ln(Earnings)$	Citations	Citations Per App	Independent Claims	Self Cite Rate
Post	-0.1429***	-0.04566***	-0.8974***	-0.2505***	-0.07455***	-0.0125***
	(0.006648)	(0.007711)	(0.2265)	(0.03112)	(0.002512)	(0.001119)
Post × Incmbnt	-0.02732***	0.1385***	-0.9082***	-0.1388***	-0.0152***	0.02579***
	(0.008251)	(0.01003)	(0.2491)	(0.02995)	(0.002722)	(0.001586)
R^2	0.3956	0.8676	0.3698	0.3067	0.2705	0.6415
N	628,000	121,000	298,000	297,000	537,000	158,000

Source: Inventor Employment History, author's calculation

Notes: Difference-in-Differences estimates for the hire event for each outcome $Y_{i,t,e}$, between inventors hired by an incumbent firm and those hired by a young firm, conditioning on post-hire records where the inventor remains at the hiring firm. Estimates are of v_1 and v_3 from Equation 5. Standard errors are shown in parentheses. Observation counts vary by measure because outcome relies a separate match, minimizing the difference in pre-match values of the given measure. Incumbent binary included but not reported. Observation counts have been rounded to avoid the disclosure of confidential information.

Hire Events Before Great Recession

Table A7: Hire Events Before Great Recession

	Citations	$ln(Earnings)$
Post	0.4137	-0.06049***
	(0.4502)	(0.01711)
Post × Incmbnt	-1.864**	0.1803***
	(0.5934)	(0.02457)
R^2	0.6801	0.9576
N	82,500	23,500

Source: Inventor Employment History, author's calculation
Notes: Estimates are of v_1 and v_3 from Equation 5 using only hires that occurred in 2005 and 2006. Standard errors are shown in parentheses. Incumbent binary included but not reported. Observation counts have been rounded to avoid the disclosure of confidential information.